

Module 7 – Big Ideas: Machine Learning (Regression & Classification)

Big Idea 1: Machine Learning = Learning Patterns to Make Predictions

- Machine learning is about using **data to learn relationships**, not hardcoding rules
- It sits inside AI and focuses on **pattern recognition + prediction**
- Two primary problem types:
 - **Regression** → **predict numbers** (price, revenue, emissions)
 - **Classification** → **predict categories** (churn, fraud, yes/no)
- The key shift from earlier modules:
 - Before: *analyze what happened*
 - Now: *predict what will happen*

Why it matters

- This is where data science becomes **decision-making and automation**
 - Every real-world ML system starts with this framing:
 - What am I predicting?
 - What data do I have?
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Big Idea 2: Regression = Modeling Continuous Outcomes

- Regression models the relationship between:
 - **Features (inputs)** → explanatory variables
 - **Target (output)** → continuous value

Core progression:

- **Simple Linear Regression**
 - One feature → one target
 - Fit a **best-fit line**
- **Multiple Linear Regression**
 - Many features → one target
 - More realistic, but risk of overfitting and collinearity

- **Polynomial / Nonlinear Regression**
 - Used when data is not a straight line
 - More flexible, but easier to overfit

Key concepts:

- Residuals = prediction error
- MSE = average squared error (how wrong your model is)

Why it matters

- Regression is everywhere:
 - Pricing models
 - Forecasting (sales, demand, weather)
 - Sports analytics (runs, performance, projections)
 - This is your **first real predictive model family**
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Big Idea 3: Classification = Predicting Probabilities and Decisions

- Classification predicts **categories**, not numbers
- Logistic regression is the core model here

Logistic regression fundamentals:

- Outputs a **probability (0 to 1)** using a sigmoid function
- Converts probability → class using a threshold (e.g., 0.5)

Key idea:

- It's still a **linear model underneath**, but transformed into probabilities

Training intuition:

- Model learns parameters that **minimize log loss**
- Uses optimization methods like **gradient descent**

Why it matters

- Most real-world business problems are classification:
 - Will a customer churn?
 - Is this fraud?

- Will a player get injured?
 - Probabilities > hard predictions
 - Enables ranking, risk scoring, and decision thresholds
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Big Idea 4: The ML Workflow Matters More Than the Model

- Machine learning is not just “fit a model”

Standard workflow:

1. Problem definition
 2. Data collection (ETL)
 3. Data preparation / EDA
 4. Model training + evaluation
 5. Deployment + monitoring
- This process is **iterative**, not linear

In practice (scikit-learn flow):

- Split data → train → predict → evaluate → repeat

Key risks:

- Overfitting (model memorizes training data)
- Collinearity (features overlap and break interpretation)
- Unrealistic “what-if” scenarios

Why it matters

- Most failures in ML are **not model issues**
 - They are:
 - bad data
 - poor problem framing
 - lack of monitoring
 - This is the bridge from:
 - notebook → production system
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Big Idea 5: Tools and Ecosystem Enable Everything

- ML is powered by an ecosystem of tools, not just algorithms

Core stack (Python):

- NumPy / Pandas → data
- Matplotlib / Seaborn → visualization
- scikit-learn → modeling

Broader ecosystem:

- Data processing (Spark, SQL)
- ML frameworks (scikit-learn, XGBoost)
- Deep learning (PyTorch, TensorFlow)
- Deployment + pipelines

Why it matters

- Knowing the workflow > memorizing algorithms
 - Real-world ML = **gluing tools together into systems**
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Big Idea 6: From Analysis to Prediction (The Career Shift)

- This module marks a major transition:

Before:

- Data cleaning
- EDA
- Visualization
- Insight generation

Now:

- Feature selection
- Model training
- Evaluation metrics
- Prediction + decision systems
- This is the difference between:
 - **Descriptive analytics** → “what happened”
 - **Predictive analytics** → “what will happen”

Why it matters

- This is where:
 - data scientists build models
 - AI engineers build systems around them